

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/02

Paper 2 Structured Questions

18 September 2013

2 hours

Candidates answer on the Question Paper
 Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
 Write in dark blue or black pen.
 You may use pencil for any diagrams, graphs or rough working.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
 A Data Booklet is provided.

You are advised to show all working in calculations.
 You are reminded of the need for good English and
 clear presentation in your answers.
 You are reminded of the need for good handwriting.
 Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [] at the end of each
 question or part question.

At the end of the examination, fasten all your work
 securely together.

For Examiner's Use	
1	12
2	15
3	12
4	9
5	13
6	11
Significant figures	
Handwriting	
Total	72

This document consists of **18** printed pages.



Answer ALL questions on the spaces provided.

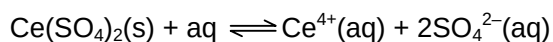
Planning (P)

- 1 The solubility of cerium(IV) sulfate, at a particular temperature, is defined as the mass of cerium(IV) sulfate that will dissolve in and just saturate 100 g of solvent at that temperature. It can also be represented by the following equation:

$$\text{Solubility} = \frac{\text{mass of cerium(IV) sulfate (g)}}{100 \text{ g of solvent}}$$

A **saturated** solution is one in which no more solid can dissolve at a particular temperature.

When cerium(IV) sulfate dissolves in water, the saturated solution is in contact with the undissolved solid to establish the following equilibrium.



Cerium(IV) sulfate is unusual. Its solubility **decreases** as the temperature of the solution increases.

Cerium(IV) sulfate crystals also dissolve in dilute sulfuric acid, H_2SO_4 , a corrosive aqueous solution.

You are to plan an experiment to investigate how the **solubility** of cerium(IV) sulfate crystals in dilute sulfuric acid depends on the concentration of the acid.

- (a) By considering the ions present in a solution of cerium(IV) sulfate in sulfuric acid, predict and explain how the solubility of the cerium(IV) sulfate will be affected by the concentration of the acid.

Prediction:

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Explanation:

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.....[1]

- (b) You are to design a two-part laboratory experiment to investigate your prediction in (a).

Part 1 – solubility of cerium(IV) sulfate in water

In addition to the standard apparatus present in a laboratory you are provided with the following materials.

100 cm³ distilled water
solid cerium(IV) sulfate, Ce(SO₄)₂

Give a step-by-step description of how you would

- prepare a saturated solution using all of the 100 cm³ of distilled water,
- control the temperature of the experiment,
- separate the saturated solution from undissolved solid,
- obtain the mass of cerium(IV) sulfate and mass of water in the saturated solution,
- calculate the solubility of cerium(IV) sulfate from the experimental results.

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Part 2 – solubility of cerium(IV) sulfate in sulfuric acid

In addition to other standard apparatus present in a laboratory, you are provided with the following materials.

100 cm³ volumetric flask
5 mol dm⁻³ sulfuric acid
distilled water
solid cerium(IV) sulfate

Give a step-by-step description of how you would prepare a range of solutions of sulfuric acid of different concentrations. These solutions could then be used to investigate the solubility of cerium(IV) sulfate in different concentrations of sulfuric acid. See part (c).

Your plan should include details of the following.

- (i) the number and concentrations of the solutions to be prepared
- (ii) the volumes of acid and water used to prepare the solutions
- (iii) the apparatus and method you would use in their preparation

You should include a table to tabulate your details for (i) and (ii).

In (i), you should prepare 5 solutions with different concentrations, and the range should cover at least a fourfold increase in concentration and at least one should be greater than 2.5 mol dm⁻³. 5 mol dm⁻³ could be one of these.

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[9]

- [1]

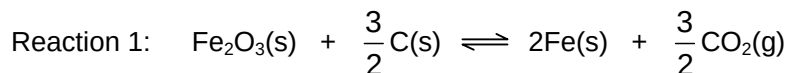
(d) State a risk that must be taken into consideration when planning the experiment.

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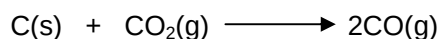
[Total: 12]

- 2 (a) Iron is commonly produced in a blast furnace from the ore haematite, which contains iron(III) oxide. Either carbon or carbon monoxide can bring about the reduction of iron(III) oxide.



	$\Delta H / \text{kJ mol}^{-1}$	$\Delta S / \text{J mol}^{-1} \text{K}^{-1}$
Reaction 1	+234	+279
Reaction 2	-24.8	+15.2

- (i) The carbon monoxide is formed by the following reaction.



By means of a suitable energy cycle or any other method, calculate the ΔH_r for this reaction using the ΔH data provided for Reactions 1 and 2.

- (ii) Calculate the ΔG of reaction 1 and 2 and state which reaction is more spontaneous at 1800 K.

- (iii) In reality, reaction 2 occurs more readily than reaction 1 at 1800 K. Suggest a reason for this.

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.....[7]

- (b) Iron and its compounds are also commonly used in organic reactions. One such reaction is the Friedel-Crafts acylation reaction. In this reaction, benzene reacts with acyl chlorides, RCOCl , in the presence of iron(III) chloride as a catalyst.

The first step of the mechanism produces the electrophile $\text{R}-\overset{\oplus}{\text{C}}=\text{O}$ that can then attack the benzene ring.

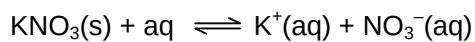
- (i) Draw the product formed when benzene reacts with ethanoyl chloride in the presence of FeCl_3 .

- (ii) Describe the mechanism of the above reaction, including the formation of the electrophile.

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[4]

- (c) (i) The dissolution of potassium nitrate in water is represented by the following equilibrium.



With the aid of the above equilibrium, explain the following phenomenon.

When potassium nitrate dissolves in water, the temperature drops. If a saturated solution of potassium nitrate at 100 °C is cooled to room temperature, crystals are formed.

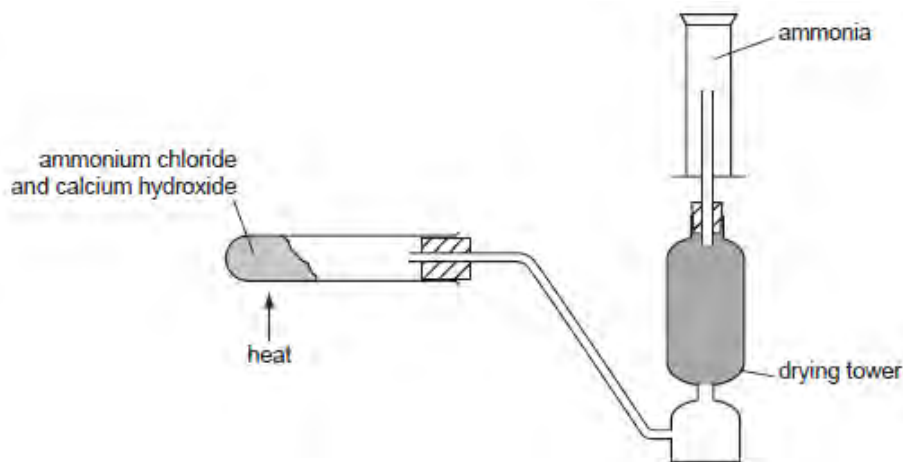
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- (ii) Write the K_c expression for the dissolution of potassium nitrate, including its units.

.....[4]

[Total: 15]

- 3 (a) Ammonia may be prepared in a school laboratory by using the apparatus below.



The reaction involves the displacement of ammonia from one of its compounds.

- (i) Construct a balanced equation for the reaction between the two reagents.

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- (ii) Common drying agents include calcium oxide, concentrated sulfuric acid and phosphorus(V) oxide.

Which of these could be used in the drying tower in this experiment? Explain your answer.

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- (b) When water is added to white anhydrous CuSO_4 , the solid dissolves to give a blue solution. On addition of concentrated NH_4Cl (aq), the solution changes to a yellow-green due to formation of copper containing species **D**. Concentrating the solution produces green crystals **E** of an ammonium salt with empirical formula $\text{CuN}_2\text{H}_8\text{Cl}_4$.

- (i) Write the electronic configuration of the copper atom.

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- (ii) Suggest the formulae of the cation present in **E**.

Cation present in **E**

- (iii) Suggest the formulae of anion **D**.

Anion **D**.....

- (iv) Suggest a balanced equation for the formation of anion **D** from aqueous CuSO_4 .

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When excess of $\text{NH}_3(\text{aq})$ is added to species **D**, the yellow green solution turns to a deep blue solution.

- (v) Use this information and the information above to suggest the strength of NH_3 , H_2O and Cl^- ligands in decreasing order.

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[5]

- (c) (i) Describe what you would see when aqueous silver nitrate is added to separate tubes containing aqueous sodium chloride, aqueous sodium bromide and aqueous sodium iodide.

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- (ii) With reference to (c)(i), describe how the relative solubilities in $\text{NH}_3(\text{aq})$ can be used to distinguish between solutions of the halide ions Cl^- , Br^- , I^- .

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[Total: 12]

- 4 (a) The solubility product, K_{sp} , of barium sulfate has a numerical value of $1.08 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$.

(i) Write an expression for the K_{sp} of barium sulfate.

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(ii) Use the value of K_{sp} given to calculate the concentration of barium sulfate in a saturated solution.

[2]

- (b) Data concerning the elements of Group II of the Periodic Table, at 298K, are given in the table. Further data can be found in the *Data Booklet*.

	Mg	Ca	Sr	Ba
$\Delta H_{\text{hydration}}$ of M^{2+} / kJ mol^{-1}	–1980	–1650	–1480	–1365
solubility of sulfate/ mol per 100 g of water	1.8×10^{-1}	4.7×10^{-3}	7.1×10^{-5}	9.4×10^{-7}

(i) When a salt such as a Group II sulfate dissolves in water, the lattice energy must be overcome. Suggest and explain how the magnitude of the lattice energy of Group II sulfates change from MgSO_4 to BaSO_4 . No calculation is required.

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(ii) Suggest why $\Delta H_{\text{hydration}}$ of M^{2+} becomes less negative from Mg^{2+} to Ba^{2+} .

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[4]

- (c) Dinitrogen pentoxide, N_2O_5 , exist as covalent molecules in the gas phase, but as ionic $[\text{NO}_2^+ \text{NO}_3^-]$ in the solid or in solution.

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Draw dot-and-cross diagrams showing the electrons (outer shells only) in NO_2^+ and NO_3^- , and predict their shapes.



[3]

[Total: 9]

- 5 Compound **W** has the molecular formula $C_8H_{11}NO$. It is an aromatic compound which contains two functional groups.

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Data about the reactions of **W** are given in the table.

reaction	reagent	result
1	Na	colourless gas evolved; white solid formed which is soluble in H_2O
2	$PCl_5(s)$	steamy white fumes
3	alkaline $I_2(aq)$	pale yellow ppt
4	$HCl(aq)$ at room temperature	colourless solution formed
5	$Br_2(aq)$ in excess	white solid formed which has $M_r = 294.8$
6	acidified $KMnO_4$ and heat	purple $KMnO_4$ decolourises

In this question, when identifying functional groups, your answers should be unambiguous.

- (a) (i) Based on reaction **1**, give the names of two different functional groups that could be present in **W**.

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- (ii) Which of the functional groups you have named in (a)(i) is confirmed by reaction **2**? Explain your answer.

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.....[4]

- (b) (i) What type of reaction takes place in reaction **3**?

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- (ii) State, as fully as you can, the functional group that reaction **3** shows to be present in **W**.

.....[2]

- (c) (i) What type of reaction takes place in reaction **4**?

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- (ii) Deduce the molecular formula of the white solid formed in reaction **5**.

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(iii) Give the functional group that reactions **4** and **5** show to be present in **W**.

.....[3]

(d) You now have enough information to determine the structural formula of **W**.

(i) Draw the fully displayed structure of **W**.

(ii) Explain clearly why you have placed **each** of the two functional groups in their particular positions.

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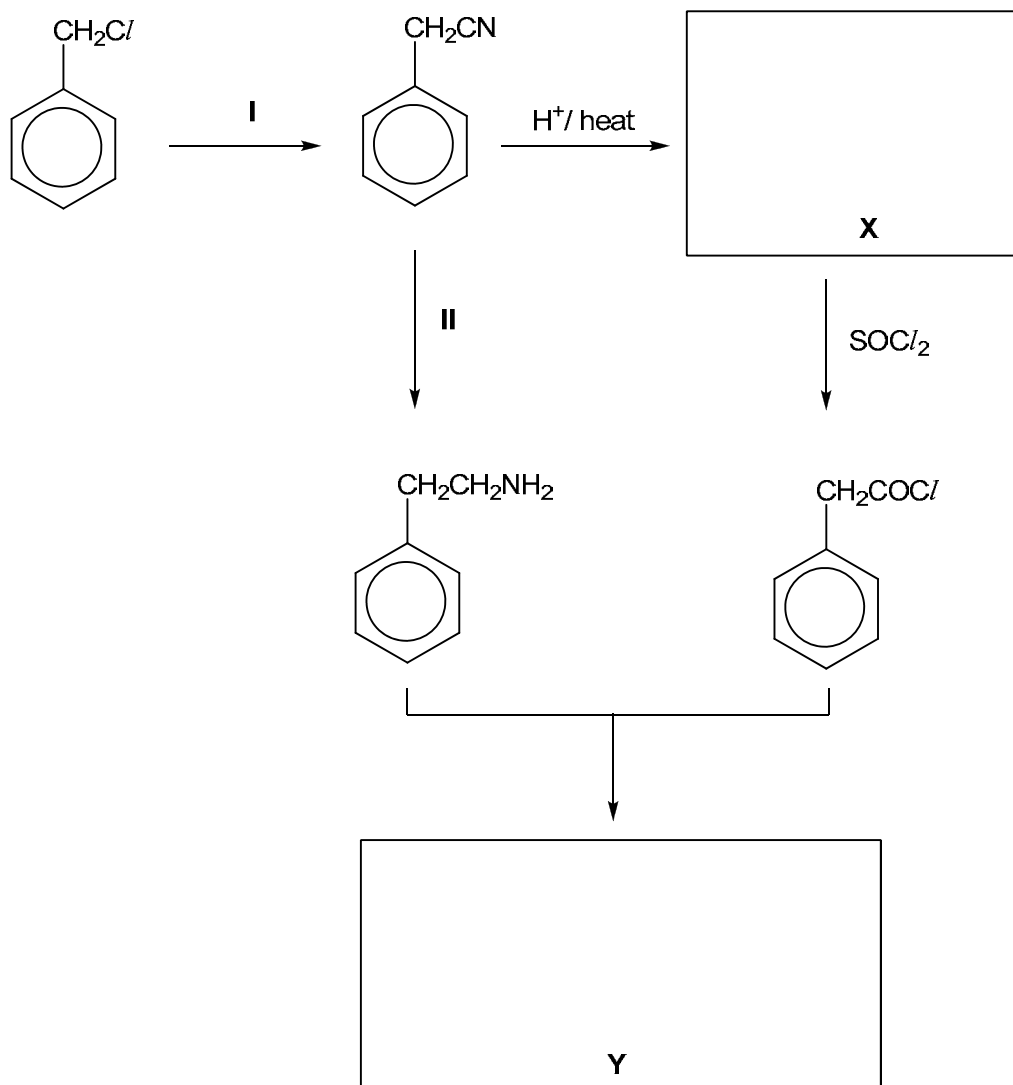
(e) Give the structural formula of the organic product of reaction **6**.

[1]

[Total: 13]

- 6 Compound **Y** is a useful intermediate in the synthesis of pharmaceuticals. **Y** can be synthesised from chloromethylbenzene according to the following scheme.

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- (a) Suggest laboratory reagents and conditions for steps **I** and **II**.

Step **I**:

Reagent:

Condition:

Step **II**:

Reagent:

Condition:

[2]

- (b) Draw the structures of **X** and **Y** in the space provided.

[2]

(c) Describe the mechanism of step I.

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[3]

(d) Suggest an explanation for the following.

When treated with warm aqueous sodium hydroxide followed by an excess of dilute nitric acid and aqueous silver nitrate, bromoethane gives a cream precipitate but CH_2CHBr gives no precipitate.

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.....[2]

- (e) Reaction of bromoethane with ethanolic ammonia produces a by-product with the formula C_2H_5NBr which gives an immediate precipitate with aqueous silver nitrate. Suggest a structure for this compound and explain why this by-product is formed.

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.....[2]

[Total: 11]